

# Diagnostic Summary Report

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## Patient Overview

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- **Patient Name:** Ashley Grahm
- **Age:** 20
- **Gender:** Female
- **Primary Complaint:** Headache and fever

## Red-Flag Alert

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- No critical red flags detected.

## Risk Stratification

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- **Level:** Medium
- **Key Drivers:**
  - Fever of 100.7 °F lasting ~24 h
  - Mild leukocytosis (WBC 10,570 cells/μL)
  - Stable vital signs (BP 125/74 mmHg, pulse 67 bpm, SpO<sub>2</sub> 98 %)
  - Absence of meningeal signs, rash, or respiratory distress

## Probable Diagnosis

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- **Condition:** Acute viral febrile illness (e.g., influenza-like illness) presenting with headache, nausea, mild leukocytosis and low-grade fever
- **Confidence:** 78 %
- **Justification (Explainability Pack):**
  - The patient reports a 24-hour history of fever (100.7 °F) and headache with associated nausea.
  - Laboratory data show mild leukocytosis (WBC 10,570), a pattern compatible with early viral infections.

- Vitals are within normal limits, reducing concern for sepsis or severe bacterial pneumonia.
- No photophobia, neck stiffness, rash, or respiratory symptoms are noted, making meningitis or severe bacterial processes less likely.
- **Reasoning:** The combination of acute, self-limited symptoms, stable vitals, and mild leukocytosis points toward a viral syndrome rather than a bacterial or more serious process.
- **Supporting Evidence:**
  - Fever 100.7 °F, duration ~24 h
  - Headache and nausea
  - Mild leukocytosis (10,570 cells/μL)
  - Stable vital signs (BP 125/74, pulse 67, SpO<sub>2</sub> 98 %)
  - No photophobia, neck stiffness, rash, or respiratory distress

## Differential Diagnoses

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- **Influenza or other viral respiratory infection** – Typical fever, headache, myalgias; mild leukocytosis may be present.
- **COVID-19 infection** – Symptom overlap; must be excluded given ongoing pandemic.
- **Acute bacterial sinusitis or upper respiratory bacterial infection** – Can cause fever and headache but usually accompanied by sinus tenderness or purulent nasal discharge (not reported).
- **Early meningitis (viral or bacterial)** – Headache and fever raise concern, but lack of neck stiffness, photophobia, altered mental status, and normal vitals make this less likely.
- **Undiagnosed type 2 diabetes mellitus (newly identified)** – Elevated fasting glucose (141 mg/dL) and HbA1c (7.1 %) suggest diabetes, which may increase infection susceptibility.
- **Vitamin B12 deficiency-related fatigue** – Low B12 (<148 pg/mL) can cause weakness and neurologic symptoms but does not explain fever.

## Key Laboratory Findings

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- **High WBC Count:** 10,570 cells/μL (reference 4,000–10,000) – mild leukocytosis.
- **High Fasting Blood Sugar:** 141 mg/dL (reference 74–106) – hyperglycemia.
- **High HbA1c:** 7.10 % (≥6.5 % diagnostic for diabetes).
- **Low Vitamin B12:** <148 pg/mL (reference 187–833) – deficiency.
- **High Homocysteine:** 23.86 μmol/L (reference 6.0–14.8) – elevated cardiovascular risk marker.

- **High IgE:** 492.30 IU/mL (reference 0-87) – suggests allergic/hypersensitivity tendency.
- **Low 25-OH Vitamin D:** 8.98 ng/mL (reference 30-100) – severe deficiency.
- **High Triglycerides:** 168 mg/dL (reference <150) – hypertriglyceridemia.
- **High Direct LDL:** 100.39 mg/dL (optimal <100) – borderline elevated.

## Recommended Plan

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### Required Investigations

- Rapid influenza antigen test or PCR – to confirm or exclude influenza.
- SARS-CoV-2 PCR or rapid antigen test – rule out COVID-19.
- Repeat CBC with differential (if symptoms evolve) – monitor leukocyte trends.
- C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR) – assess inflammatory burden.
- Comprehensive metabolic panel – evaluate electrolytes, renal and hepatic function.
- Urinalysis and urine culture – exclude urinary source of infection.
- Chest X-ray (if cough or respiratory symptoms develop) – screen for pneumonia.
- Blood cultures (if fever persists >48 h or clinical deterioration) – detect bacteremia.
- Repeat fasting glucose and HbA1c in 2-3 months – confirm diabetes diagnosis.
- Serum vitamin B12 repeat and methylmalonic acid – verify deficiency and guide supplementation.
- Serum 25-OH vitamin D repeat after supplementation – monitor response.
- Lipid panel follow-up in 3-6 months – track dyslipidemia.
- Lumbar puncture with CSF analysis (if meningeal signs appear) – rule out meningitis.

### Suggested Medications / Treatments

- **Acetaminophen 500 mg PO q6h PRN** for fever/pain (max 3 g/24 h).
- **Ondansetron 4 mg PO q8h PRN** for nausea, if needed.
- Encourage **oral hydration (2-3 L/day)** and rest.
- If bacterial sinusitis becomes likely: **Amoxicillin 500 mg PO TID for 5-7 days**.
- If diabetes confirmed: lifestyle counseling; consider **Metformin 500 mg PO BID** after repeat labs & renal assessment.

- **Vitamin B12 supplementation:** Cyanocobalamin 1000 µg PO daily × 1 month, then reassess.
- **Vitamin D supplementation:** Cholecalciferol 2000 IU PO daily (adjust per follow-up levels).
- If IgE-mediated allergy suspected: **Cetirizine 10 mg PO daily.**

**Medication Disclaimer:** A qualified human doctor must make the final prescribing decision, taking into account the full clinical picture, allergies, contraindications, and local guidelines.

## Information Gaps

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- Exact onset time of headache and fever (hours vs. days).
- Presence or absence of respiratory symptoms (cough, sore throat, nasal congestion).
- Detailed physical exam findings for photophobia, neck stiffness, or other meningeal signs.
- Recent travel, sick contacts, or COVID-19 exposure history.
- Menstrual history and pregnancy status (relevant for medication safety).
- Current use of antipyretics or analgesics prior to presentation.